LAPORAN WORKSHOP DEVELOPER OPERATIONAL

MINGGU – 11

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2026

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BAB I

MINGGU 11 (Kubernetes Basic Setup)

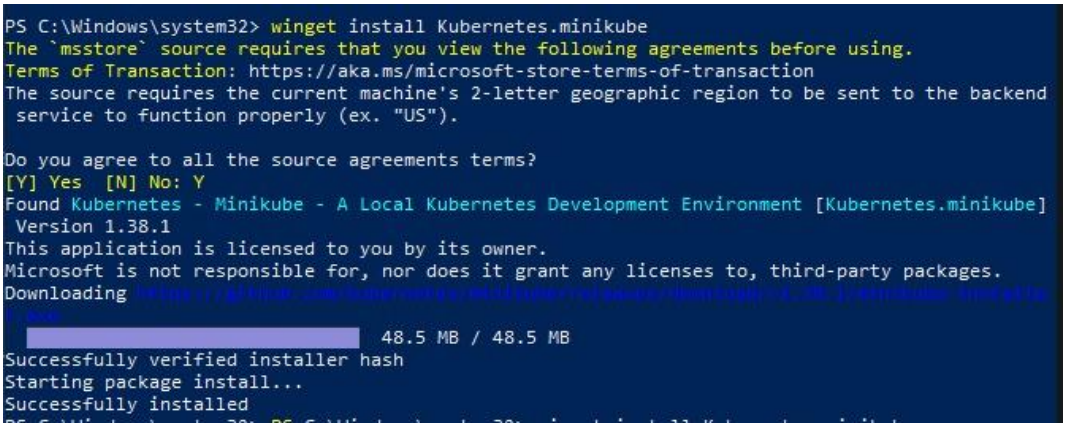
1.1 Prosedur Kerja

Instalasi Kubernates & melakukan basic setup.

1.2 Hasil dan Pembahasan

1. Setup Kubernates di Windows

   Pada tahapan ini dilakukan instalasi Minikube di Windows dengan menggunakan perintah winget install Kubernetes.minikube melalui PowerShell.



Gambar 1. 1 Install Minikube

```
PS C:\Windows\system32> minikube start --driver=docker
* minikube v1.38.1 on Microsoft Windows 10 Pro 22H2
* Using the docker driver based on user configuration
! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for mo
re info.
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Downloading Kubernetes v1.35.1 preload ...
    > preloaded-images-k8s-v18-v1...:  239.25 MiB / 272.45 MiB  87.81% 62.92 Ki
    > gcr.io/k8s-minikube/kicbase...:  262.27 MiB / 519.58 MiB  50.48% 68.93 Ki
    > index.docker.io/kicbase/sta...:  519.58 MiB / 519.58 MiB  100.00% 493.81
! minikube was unable to download gcr.io/k8s-minikube/kicbase:v0.0.50, but successfully downlo
aded docker.io/kicbase/stable:v0.0.50@sha256:eb4fec00e8ad70adf8e6436f195cc429825ffb85f95afcdb5
d8d9deb576f3e93 as a fallback image
* Creating docker container (CPUs=2, Memory=3072MB) ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...

X Exiting due to INET_DOWNLOAD_TIMEOUT: update primary control-plane node: downloading binarie
s: downloading kubectl: download failed: https://dl.k8s.io/release/v1.35.1/bin/linux/amd64/kub
ectl?checksum=file:https://dl.k8s.io/release/v1.35.1/bin/linux/amd64/kubectl.sha256: getter: &
{Ctx:context.Background Src:https://dl.k8s.io/release/v1.35.1/bin/linux/amd64/kubectl?checksum
=file:https://dl.k8s.io/release/v1.35.1/bin/linux/amd64/kubectl.sha256 Dst:C:\Users\T430S\.min
ikube\cache\linux\amd64\v1.35.1/kubectl.download Pwd: Mode:2 Umask:---------- Detectors:[0x7ff
624482400 0x7ff624482400 0x7ff624482400 0x7ff624482400 0x7ff624482400 0x7ff624482400 0x7ff6244
82400] Decompressors:map[bz2:0xc0002eff48 gz:0xc0003d8000 tar:0xc0002eff80 tar.bz2:0xc0002effb
0 tar.gz:0xc0002effc0 tar.xz:0xc0002effe0 tar.zst:0xc0002efff0 tbz2:0xc0002effb0 tgz:0xc0002ef
fc0 txz:0xc0002effe0 tzst:0xc0002efff0 xz:0xc0003d8008 zip:0xc0003d8010 zst:0xc0003d8020] Gett
ers:map[file:0xc0021ada90 http:0xc0006c4640 https:0xc0006c4690] Dir:false ProgressListener:0x7
ff62442bef0 Insecure:false DisableSymlinks:false Options:[0x7ff61fc660c0]}: invalid checksum:
Error downloading checksum file: Get "https://dl.k8s.io/release/v1.35.1/bin/linux/amd64/kubect
l.sha256": net/http: TLS handshake timeout
* Suggestion: A firewall is likely blocking minikube from reaching the internet. You may need
to configure minikube to use a proxy.
* Documentation: https://minikube.sigs.k8s.io/docs/handbook/vpn_and_proxy/
* Related issue: https://github.com/kubernetes/minikube/issues/3846

PS C:\Windows\system32> kubectl get nodes
E0429 09:13:17.895859    8760 memcache.go:265] "Unhandled Error" err="couldn't get current ser
ver API group list: the server could not find the requested resource"
E0429 09:13:18.063989    8760 memcache.go:265] "Unhandled Error" err="couldn't get current ser
ver API group list: the server could not find the requested resource"
E0429 09:13:18.082529    8760 memcache.go:265] "Unhandled Error" err="couldn't get current ser
ver API group list: the server could not find the requested resource"
```

Gambar 1. 2 Kendala saat Menjalankan Minikube

Setelah proses install, selanjutnya adalah menjalankan minkube. Pada bagian ini saya mengalami kendala internet, mencoba beberapa kali diulang hasilnya tetap sama. Namun akhirnya setelah beberapa lama berhasil dijalankan. Hal ini dibuktikan saat menjalankan perintah “kubectl get node” hasilnya adalah ready.

```
PS C:\Windows\system32> minikube start --driver=docker
* minikube v1.38.1 on Microsoft Windows 10 Pro 22H2
* Using the docker driver based on existing profile
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Downloading Kubernetes v1.35.1 preload ...
    > preloaded-images-k8s-v18-v1...:  272.45 MiB / 272.45 MiB  100.00% 1.91 Mi
! Executing "docker container inspect minikube --format={{.State.Status}}" took an unusually l
ong time: 5.6485847s
* Restarting the docker service may improve performance.
* Updating the running docker "minikube" container ...
    > kubelet.sha256:  64 B / 64 B [-------------------------] 100.00% ? p/s 0s
    > kubeadm.sha256:  64 B / 64 B [-------------------------] 100.00% ? p/s 0s
    > kubectl.sha256:  64 B / 64 B [-------------------------] 100.00% ? p/s 0s
    > kubelet:  55.42 MiB / 55.42 MiB [-----------] 100.00% 95.42 KiB p/s 9m55s
    > kubectl:  55.88 MiB / 55.88 MiB [-----------] 100.00% 94.86 KiB p/s 10m3s
    > kubeadm:  69.02 MiB / 69.02 MiB [---------] 100.00% 108.76 KiB p/s 10m50s
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Configuring bridge CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: default-storageclass, storage-provisioner
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
PS C:\Windows\system32> kubectl get nodes
NAME       STATUS   ROLES           AGE     VERSION
minikube   Ready    control-plane   3m18s   v1.35.1
```

Gambar 1. 3 Menjalankan Minikube

2. Membuat Folder Project

   Setelah berhasil menjalankan, selanjutnya adalah membuat folder project, yang nantinya didalam folder tersebut terdapat file deployment.yaml dan service.yaml. Berikut Adalah dokumentasinya.

```
PS D:\sem6\project> mkdir kubernetes


    Directory: D:\sem6\project


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         4/29/2026   9:42 AM                kubernetes


PS D:\sem6\project> cd kubernetes
PS D:\sem6\project\kubernetes> code .
```

Gambar 1. 4 Membuat Folder Project

```yaml
! deployment.yaml
1   apiVersion: apps/v1
2   kind: Deployment
3   metadata:
4     name: web-nginx
5   spec:
6     replicas: 3
7     selector:
8       matchLabels:
9         app: web-nginx
10    template:
11      metadata:
12        labels:
13          app: web-nginx
14      spec:
15        containers:
16        - name: nginx
17          image: nginx
18          ports:
19          - containerPort: 80
```

Gambar 1. 5 Isi pada File deployment.yaml

```yaml
! service.yaml
1   apiVersion: v1
2   kind: Service
3   metadata:
4     name: web-nginx-service
5   spec:
6     selector:
7       app: web-nginx
8     ports:
9       - port: 80
10        targetPort: 80
11    type: NodePort
```

Gambar 1. 6 Isi pada File service.yaml

3. Menjalankan YAML

   Setelah kedua file yaml terisi, selanjutnya adalah melakukan apply file dengan menjalankan peintah “kubectl apply -f deployment.yaml” dan “kubectl apply -f service.yaml”. Kemudian setelah itu lihat semua resource dengan perintah “kubectl get all”. Selanjutnya cek pod dengan menjalankan perintah “kubectl get pods”, dan terakhir melakukan monitoring live dengan perintah “kubectl get pods -w”.

```
PS D:\sem6\project\kubernetes> kubectl apply -f deployment.yaml
>> kubectl apply -f service.yaml
deployment.apps/web-nginx created
service/web-nginx-service created
PS D:\sem6\project\kubernetes> kubectl get all
NAME                              READY   STATUS              RESTARTS   AGE
pod/web-nginx-748d764f7-jd87c     0/1     ContainerCreating   0          15s
pod/web-nginx-748d764f7-kcmhs     0/1     ContainerCreating   0          15s

NAME                      TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
service/kubernetes        ClusterIP   10.96.0.1       <none>        443/TCP        16m

NAME                      READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/web-nginx 0/3     3            0           15s

NAME                                 DESIRED   CURRENT   READY   AGE
replicaset.apps/web-nginx-748d764f7  3         3         0       15s
PS D:\sem6\project\kubernetes> kubectl get pods
NAME                      READY   STATUS              RESTARTS   AGE
web-nginx-748d764f7-jd87c 0/1     ContainerCreating   0          45s
web-nginx-748d764f7-jsx2q 0/1     ContainerCreating   0          45s
web-nginx-748d764f7-kcmhs 0/1     ContainerCreating   0          45s
PS D:\sem6\project\kubernetes> kubectl get pods -w
NAME                      READY   STATUS              RESTARTS   AGE
web-nginx-748d764f7-jd87c 0/1     ContainerCreating   0          59s
web-nginx-748d764f7-jsx2q 0/1     ContainerCreating   0          59s
web-nginx-748d764f7-kcmhs 0/1     ContainerCreating   0          59s
web-nginx-748d764f7-jsx2q 1/1     Running             0          81s
web-nginx-748d764f7-jd87c 1/1     Running             0          84s
web-nginx-748d764f7-kcmhs 1/1     Running             0          88s
```

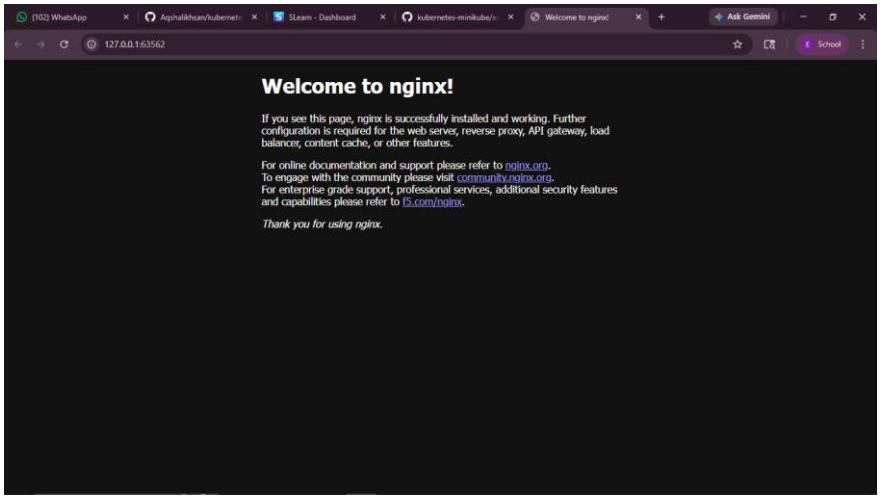
Gambar 1. 7 Menjalankan YAML

4. Membuka Aplikasi

   Selanjutkan membuka aplikasi dengan menggunakan IP yang muncul.

```
PS D:\sem6\project\kubernetes> minikube service web-nginx-service
|-----------|-----------------|-------------|---------------------------|
| NAMESPACE |      NAME       | TARGET PORT |           URL             |
|-----------|-----------------|-------------|---------------------------|
| default   | web-nginx-service | 80        | http://192.168.49.2:31872 |
|-----------|-----------------|-------------|---------------------------|
🔗  Starting tunnel for service web-nginx-service.
|-----------|-----------------|-------------|-----------------------|
| NAMESPACE |      NAME       | TARGET PORT |          URL          |
|-----------|-----------------|-------------|-----------------------|
| default   | web-nginx-service |           | http://127.0.0.1:63562 |
|-----------|-----------------|-------------|-----------------------|
🎉  Opening service default/web-nginx-service in default browser...
❗  Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```

Gambar 1. 8 Melihat IP



Gambar 1. 9 Tampilan Setelah Dijalankan

5. Scaling Pod

   Pada terminal 1 menjalankan perintah “kubectl get pods -w” untuk melihat pod live.

```
PS D:\sem6\project\kubernetes> kubectl get pods -w
NAME                    READY   STATUS    RESTARTS   AGE
web-nginx-748d764f7-jd87c   1/1     Running   0          23m
web-nginx-748d764f7-jsx2q   1/1     Running   0          23m
web-nginx-748d764f7-kcmhs   1/1     Running   0          23m
web-nginx-748d764f7-fqfsg   0/1     Pending   0          0s
web-nginx-748d764f7-95f5v   0/1     Pending   0          0s
web-nginx-748d764f7-95f5v   0/1     Pending   0          0s
web-nginx-748d764f7-fqfsg   0/1     Pending   0          0s
web-nginx-748d764f7-95f5v   0/1     ContainerCreating   0          1s
web-nginx-748d764f7-fqfsg   0/1     ContainerCreating   0          1s
web-nginx-748d764f7-95f5v   1/1     Running             0          8s
web-nginx-748d764f7-fqfsg   1/1     Running             0          13s
```

Gambar 1. 10 Lihat Pod Live

```
PS D:\sem6\project\kubernetes> kubectl scale deployment web-nginx --replicas=5
deployment.apps/web-nginx scaled
```

Gambar 1. 11 Scale Up ke 5

```
PS D:\sem6\project\kubernetes> kubectl scale deployment web-nginx --replicas=3
deployment.apps/web-nginx scaled
```

Gambar 1. 12 Scale Down ke 3

```
PS D:\sem6\project\kubernetes> kubectl get pods -w
NAME                          READY   STATUS    RESTARTS   AGE
web-nginx-748d764f7-jd87c     1/1     Running   0          23m
web-nginx-748d764f7-jsx2q     1/1     Running   0          23m
web-nginx-748d764f7-kcmhs     1/1     Running   0          23m
web-nginx-748d764f7-fqfsg     0/1     Pending   0          0s
web-nginx-748d764f7-95f5v     0/1     Pending   0          0s
web-nginx-748d764f7-95f5v     0/1     Pending   0          0s
web-nginx-748d764f7-fqfsg     0/1     Pending   0          0s
web-nginx-748d764f7-95f5v     0/1     ContainerCreating   0          1s
web-nginx-748d764f7-fqfsg     0/1     ContainerCreating   0          1s
web-nginx-748d764f7-95f5v     1/1     Running             0          8s
web-nginx-748d764f7-fqfsg     1/1     Running             0          13s
web-nginx-748d764f7-fqfsg     1/1     Terminating         0          103s
web-nginx-748d764f7-95f5v     1/1     Terminating         0          103s
web-nginx-748d764f7-fqfsg     1/1     Terminating         0          103s
web-nginx-748d764f7-95f5v     1/1     Terminating         0          104s
web-nginx-748d764f7-fqfsg     0/1     Completed           0          105s
web-nginx-748d764f7-95f5v     0/1     Completed           0          105s
web-nginx-748d764f7-95f5v     0/1     Completed           0          106s
web-nginx-748d764f7-95f5v     0/1     Completed           0          106s
web-nginx-748d764f7-95f5v     0/1     Completed           0          106s
web-nginx-748d764f7-fqfsg     0/1     Completed           0          106s
web-nginx-748d764f7-fqfsg     0/1     Completed           0          106s
web-nginx-748d764f7-fqfsg     0/1     Completed           0          106s
```

Gambar 1. 13 Deployment Live pada Terminal Lain

6. Menambah Worker Node

```
PS D:\sem6\project\kubernetes> minikube node add
😄  Adding node m02 to cluster minikube as [worker]
❗  Cluster was created without any CNI, adding a node to it might cause broken networking.
👍  Starting "minikube-m02" worker node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
💾  Downloading Kubernetes v1.35.1 preload ...
🔥  Creating docker container (CPUs=2, Memory=2200MB) ...
🐳  Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔎  Verifying Kubernetes components...
🌟  Successfully added m02 to minikube!
PS D:\sem6\project\kubernetes> kubectl get nodes
NAME           STATUS     ROLES           AGE    VERSION
minikube       Ready      control-plane   49m    v1.35.1
minikube-m02   NotReady   <none>          2m3s   v1.35.1
PS D:\sem6\project\kubernetes> minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

minikube-m02
type: Worker
host: Running
kubelet: Running

PS D:\sem6\project\kubernetes>
```

Gambar 1. 14 Menambah Worker Node